

Advanced (Carolina Reaper)

Open Ended Questions (FRQ)

Q9. A person has $\frac{1}{2}$ cup of oil and $\frac{3}{4}$ cup of vinegar. They want to mix the oil and vinegar in a 2 : 3 ratio. How much more oil and vinegar do they need to make the mixture? Show your work and explain your reasoning.

Q10. A water tank can hold 1200 gallons of water. If $\frac{1}{3}$ of the tank is already filled, and $\frac{1}{4}$ of the tank is filled every hour, how many hours will it take to fill the rest of the tank? Show your work and explain your reasoning.

Chef's Tips

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- Tip 1: Make sure to check the units of measurement in each question.
- Tip 2: Encourage students to use visual aids such as number lines or graphs to help solve the problems.
- Tip 3: Remind students to show their work and explain their reasoning for the free-response questions.